

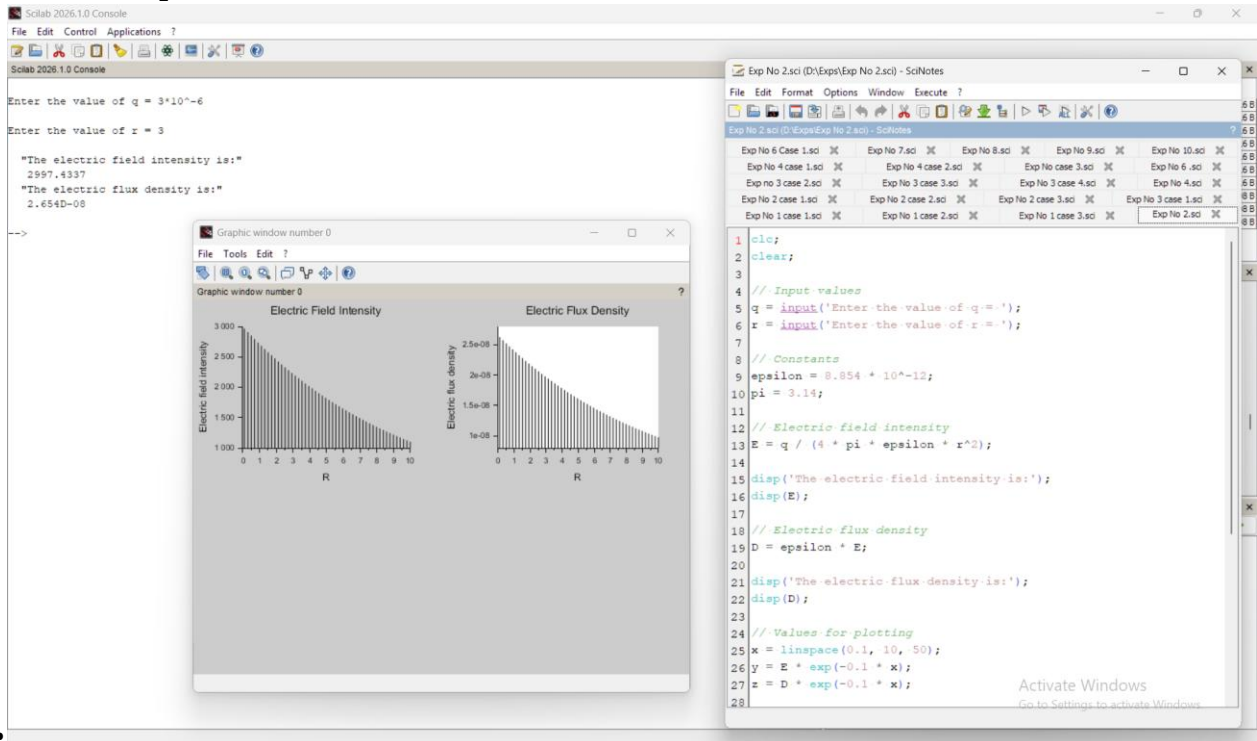
Exp. No: 2

Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim:

To write program for electric field and electric flux density in sphere using SCILAB.

Software required:



SCILAB version 6.0.

EXP 2:-

$$q = 3 \times 10^{-6}, r = 3 \text{ m}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

electric field $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$

O/P ✓ $E = \frac{9 \times 10^9 \times 3 \times 10^{-6}}{3^2}$

$$E = \frac{27 \times 10^5}{9} = \boxed{2997.33 \text{ N/C}}$$

electric flux density

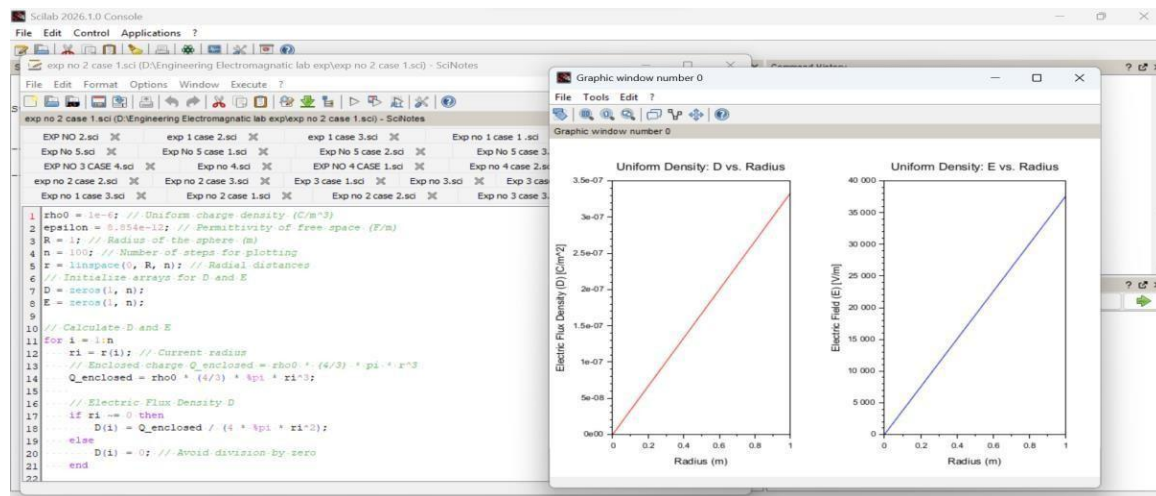
$$D = \epsilon_0 E$$

$$= (8.854 \times 10^{-12}) (2997.33)$$

$$\boxed{D = 2.654 \times 10^{-8} \text{ C/m}^2}$$

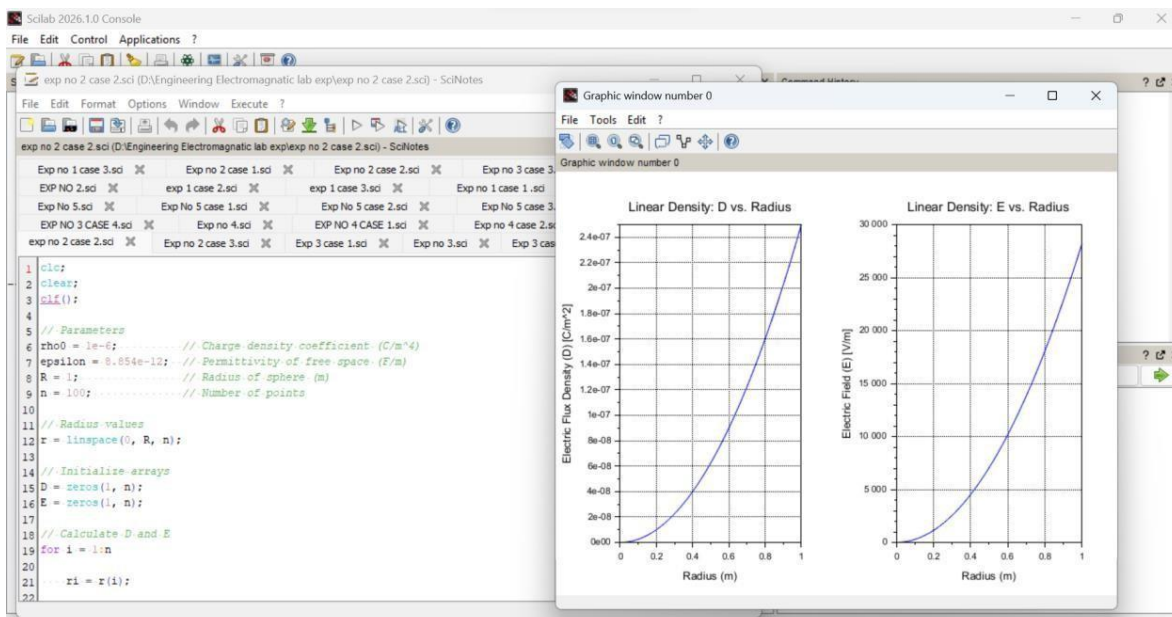
Case 1:

Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0 . What are the values of E and D at the center and at the surface of the sphere?



Case 2:

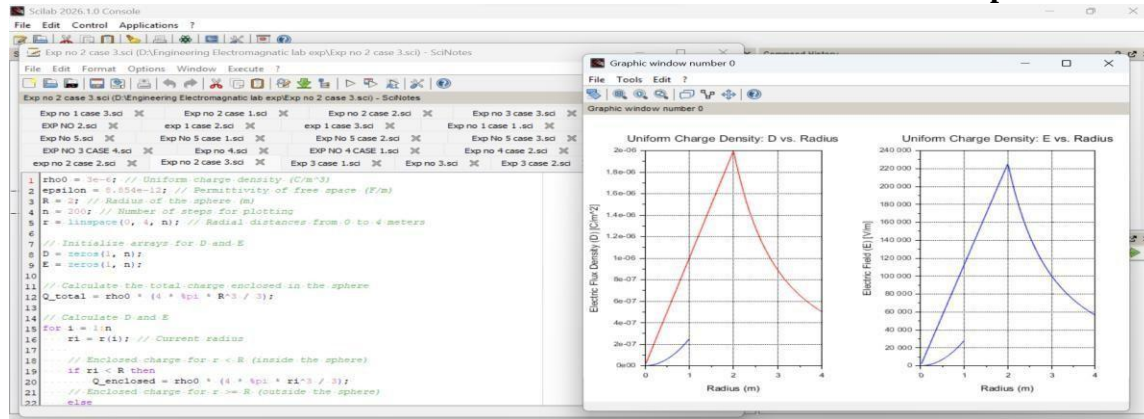
For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive the expressions for the electric field (E) and electric flux density (D). Plot their variations inside the sphere and analyze the trends.



Case 3:

A sphere of radius $R=2$ m has a uniform charge density $\rho=3 \times 10^{-6}$ C/m³

1. Calculate and plot the electric flux density $D(r)$ and electric field $E(r)$ as a function of radial distance r , ranging from 0 to 4 meters.
2. Discuss the behaviour of the electric field inside and outside the sphere.



Result:

Thus, the program was verified for electric field and electric flux density in sphere using SCILAB was successfully.